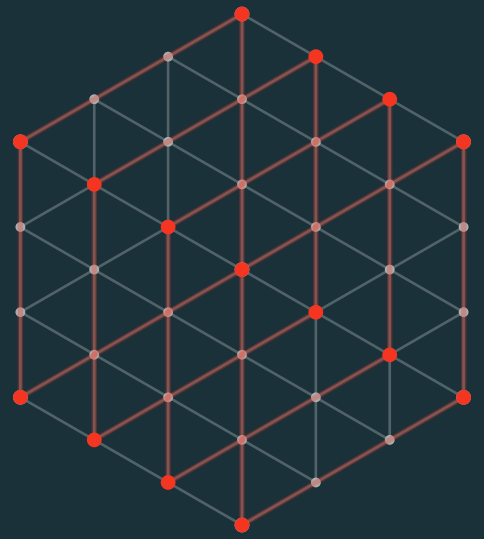
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DIGITAL NATIVE COMMERCE



Real-time personalization on Databricks

The right product for every shopper within about 100 ms of a click, governed, explained, and measured on one platform.

AGENDA

From the problem to a pilot plan

01 The business challenge

02 The Databricks solution and architecture

03 Outcomes from the A/B experiment

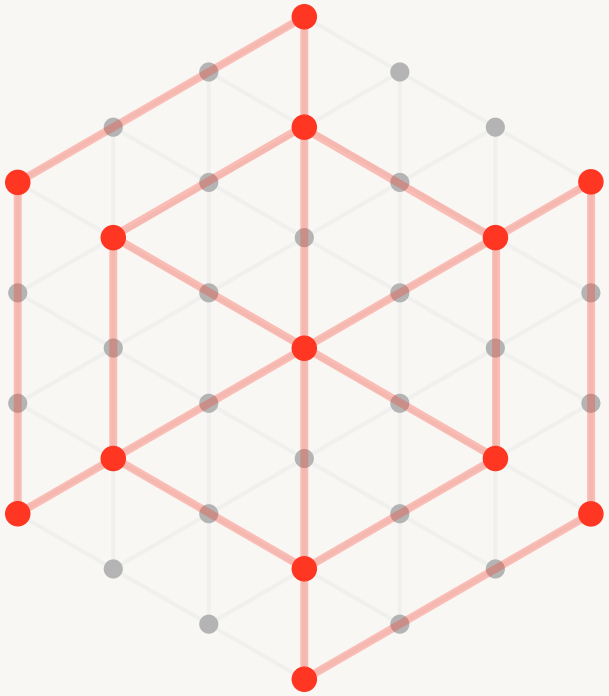
04 How to get started

THE BUSINESS CHALLENGE

Paid traffic lands on a *generic* store

A first-time visitor and a loyal repeat buyer see the same homepage, rails, and offers.

- Low session-to-purchase conversion and high bounce on landing pages
- About 30% of sessions are cold start, with no history to personalize on
- Merchandisers can't see why a product was shown, or whether it worked
- Personalization logic lives in app code, away from the analytics the business trusts



WHY IT IS HARD TODAY

A stitched stack makes every step a hand-off

Need	Stitched stack	Databricks
Fresh signals	Nightly batch features	Last 5 clicks read from Lakebase per request
Serving data	Separate database, custom ETL	Gold tables synced to Lakebase automatically
Governance	Rules re-built per system	Unity Catalog masks, filters, lineage
Explainability	None for the business	Grounded "why this?" for every recommendation
Insight	Analyst ticket, hours	Genie and a copilot, seconds

SECTION 02

The Databricks solution, from click to recommendation



THE SOLUTION

Three capabilities on one governed platform

Real-time signal

Lakeflow streams clickstream into governed tables. Lakebase serves features and live session state in milliseconds.

Intelligence

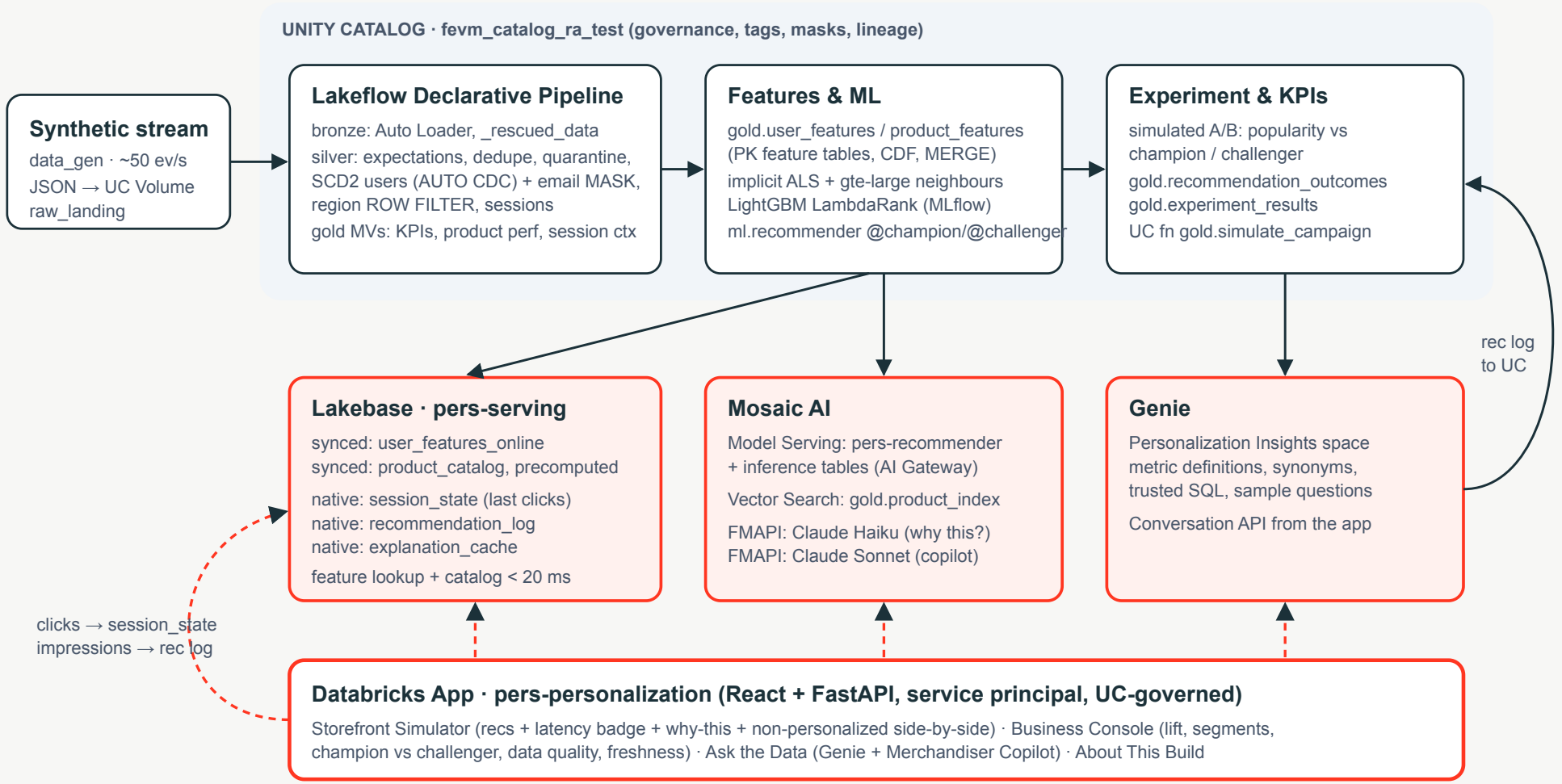
A two-stage recommender on Model Serving, Vector Search for cold start, and grounded LLM explanations.

Business loop

An A/B experiment, a Genie space, and a merchandiser copilot that simulates campaigns.

ARCHITECTURE

Every stage consumes the previous stage's output



HOW IT WORKS

Four steps turn a click into the next recommendation

1

Capture the click

The app writes each view, add-to-cart, and purchase to Lakebase session state.

2

Look up context

User features and the last 5 clicks come back from Lakebase in about 5 ms.

3

Rank candidates

ALS and embedding candidates are ranked by LightGBM on Model Serving.

4

Explain and log

Claude writes a grounded "why this?" and every impression is logged for attribution.

BUILT FOR PRODUCTION

Governed, resilient, and explainable by design

01

Governed end to end

Email masked and tagged as PII, regional row filters, and lineage from landing volume to served model in Unity Catalog.

02

Resilient serving

If Model Serving is slow, precomputed recommendations in Lakebase fill the rail, so shoppers never see an empty page.

03

Explainable and measured

Champion and challenger models in Unity Catalog, an A/B readout, and a copilot scored 20 of 20 on grounded answers.

SECTION 03

Outcomes from a 14-day A/B experiment

PERSONALIZED VS POPULARITY BASELINE

Personalization doubles engagement

2x

recommendation CTR, 3.1% to 6.1% for returning shoppers

+47%

session conversion, 1.93% to 2.84% with the champion model

2x

cold-start conversion, 0.83% to 1.67% for brand-new shoppers

Simulated A/B on synthetic data; outcomes drawn from hidden shopper preferences the model never sees

SCORECARD AGAINST TARGETS

Engagement targets are met; AOV and latency are the next focus

Metric	Baseline	Personalized	Target
Recommendation CTR	3.1%	**6.1%**	at least 6%
Session conversion	1.93%	**2.84% to 3.30%**	at least 2.6%
Cold-start conversion	0.83%	**1.67%**	at least 1.4%
Average order value	$57.92	**$56.47 to $60.34**	plus 5 to 8%
Recommendation latency	n/a	**about 112 ms p50**	under 100 ms p95

Returning shoppers unless noted. Range shows champion to challenger. Offline NDCG@10 rose from 0.072 to 0.121.

GET STARTED

One bundle deploy and one setup job

- Everything is a Declarative Automation Bundle on serverless compute
- Swap the synthetic generator for your own clickstream feed
- Keep the pipeline, models, Lakebase, Genie, and app as they are

```
# deploy pipeline, jobs, Lakebase, app
databricks bundle deploy

# data, features, models, serving, Genie
databricks bundle run pers_setup
databricks bundle run pers_app
```

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Thank you

Next step: a two-week pilot on your own clickstream.

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